WORKSHEET - NAPLAN Year 9

Number and Algebra

Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-95621

A buoy is in the ocean recording the waves passing by.

At sea level it measures 0 m. It moves above and below sea level.

Which list gives four sea level recordings arranged in order lowest to highest?

- 0 m, –2 m, 4 m,–4 m
- –2 m, –5 m, 0 m, 1 m
- –5 m, 2 m, –3 m, 0 m
- –5 m, –3 m, 0 m, 2 m

2. Number and Algebra, NAP-83792

$7 \times 2^3 =$

28 42 56 2744

3. Number and Algebra, NAP-01565

$\sqrt{190}$ is between

$9 \text{ and } 12$ $12 \text{ and } 14$ $14 \text{ and } 20$ $40 \text{ and } 50$

4. Number and Algebra, NAP-36749

12 people went to Sea World together in a bus.

Entry into Sea World was $44 per person and the total cost of the bus was $120.

Which expression shows the total cost of the trip for everybody?

$(44+120) \times 12$ $44+12+120$ $(12 \times 44)+120$ 12×44

5. Number and Algebra, NAP-13258

The Olympic Games are held every 4 years.

The 30th Games were held in London in 2012.

What year will the 38th Olympics be held?

2020 2038 2040 2044

6. Number and Algebra, NAP-66218

Peter thinks of a number, n.

He triples the number.

He then halves that answer and subtracts 25.

Which expression shows what Peter did?

$\frac{3n-25}{2}$ $\frac{n^3}{2}-25$ $\frac{3n}{2}-25$ $\frac{n^3-25}{2}$

7. Number and Algebra, NAP-66248

The age of the Universe is estimated to be 14 000 000 000 years old.

This number can also be written as

1.4×10^9 1.4×10^{10} 14×10^{10} 14×10^{11}

8. Number and Algebra, NAP-19207

Pamela has these four number cards.

3^4 4^3 6^2 9^2

Which two cards have the same value?

- 3^4 and 4^3
- 9^2 and 4^3
- 3^4 and 6^2
- 9^2 and 3^4

Worked Solutions

1. Number and Algebra, NAP-95621

-5 m, -3 m, 0 m, 2 m

2. Number and Algebra, NAP-83792

$$\begin{aligned} 7 \times 2^3 &= 7 \times 2 \times 2 \times 2 \\ &= 56 \end{aligned}$$

3. Number and Algebra, NAP-01565

$$12^2 = 144$$
$$13^2 = 169$$
$$14^2 = 196$$

$\therefore \sqrt{190}$ is between 12 and 14.

4. Number and Algebra, NAP-36749

Total cost of entry to Sea World

$= 12 \times \$44$

Total cost of bus $= \$120$ (given)

$\therefore$ Total cost of the trip for everybody

$= (12 \times 44) + 120$

5. Number and Algebra, NAP-13258

$$\begin{aligned} \text{38th Olympics date} &= 2012 + (8 \times 4) \\ &= 2044 \end{aligned}$$

6. Number and Algebra, NAP-66218

Triples the number: $n \rightarrow 3n$

Halves the answer: $3n \rightarrow \dfrac{3n}{2}$

Deducts 25: $\dfrac{3n}{2} \rightarrow \dfrac{3n}{2} - 25$

7. Number and Algebra, NAP-66248

1.4×10^{10}

8. Number and Algebra, NAP-19207

$$\begin{aligned} 9^2 &= 9 \times 9 \\ &= 81 \\ 3^4 &= 3 \times 3 \times 3 \times 3 \\ &= 81 \end{aligned}$$

$\therefore\ 9^2$ and 3^4 have the same value.`